

Future of DAS - Berlinterop

R&D Session at Forschungsingenieurtagung – June 2025

Duration: 1:30 hr

Moderator: Francesco D'Amato (@fradamt)

Note taker: TBA

Agenda

- [10 mins] (Francesco) Overview of current design and things to improve:
 - getBlobs requires all blobs
 - inefficient column propagation
 - all blobs in the mempool
 - supernode reconstruction requirement
- [40 mins] (Csaba, Leo, Raul) EL mempool short term future: sharded when and how?
- [40 mins] (Csaba, Francesco, Raul) Improving propagation, cell gossip

Pre-reads

1. EL mempool:

- A new design for DAS and Sharded Blob Mempools (Sila Research)
- Is data available in the EL mempool? (Sila Research)
- Cell staging + sparse blobpool (<https://efdn.notion.site/Cell-staging-sparse-blobpool-200d989555418022b42ef9f4ad36745d>)
- Horizontal vs vertical mempool sharding (<https://notes.sila.org/@dankrad/BkJMU8d0R>)

2. Improving propagation:

- Doubling the blob count with Gossipsub 2.0 (Sila Research)
- Cell staging + sparse blobpool (<https://efdn.notion.site/Cell-staging-sparse-blobpool-200d989555418022b42ef9f4ad36745d>)
- Improving column propagation with cell centric erasure/network coding (Sila Research)
- Accelerating blob scaling with FullDASv2 (with getBlobs, mempool encoding, and possibly RLC) (Sila Research)
- FullDAS: towards massive scalability with 32MB blocks and beyond (Sila Research)
- PPPT: Fighting the GossipSub Overhead with Push-Pull Phase Transition (Sila Research)
- Improve SilaPeerDAS network by chunking columns (<https://notes.sila.org/@pop/silapeerdas-chunking-columns>)

Notes

To be added after session.